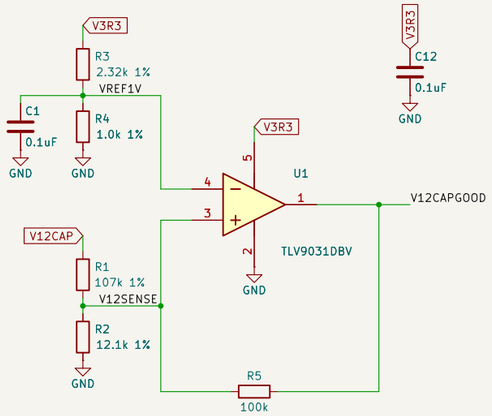
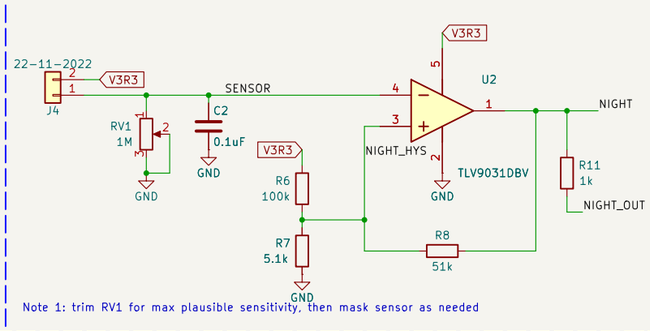
V12CAP Monitor (U1)

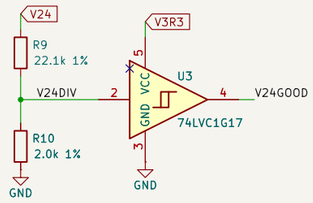


Night Sensor (U2, J4)

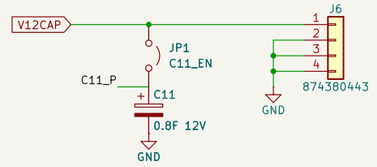


Note 1: trim RV1 for max plausible sensitivity, then mask sensor as needed

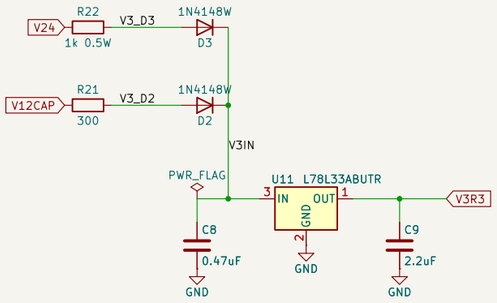
V24 Detect (U3)



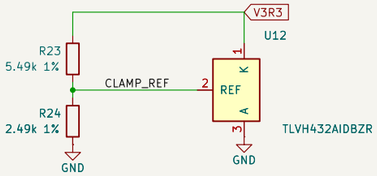
Backup Capacitor (C11, J6)



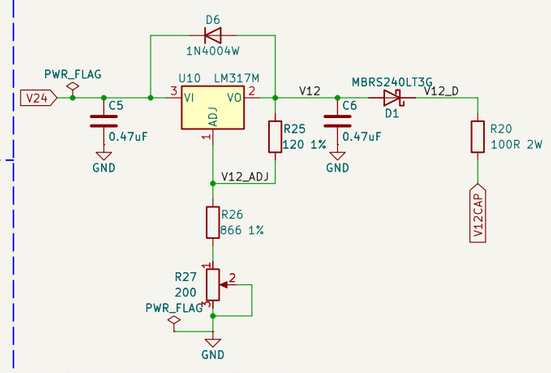
3V3 Regulator (U11)



3V3 Clamp (U12)

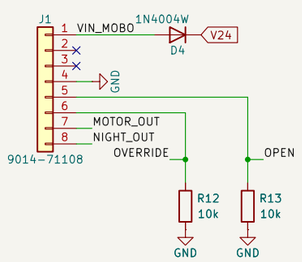


12V Regulator (U10)

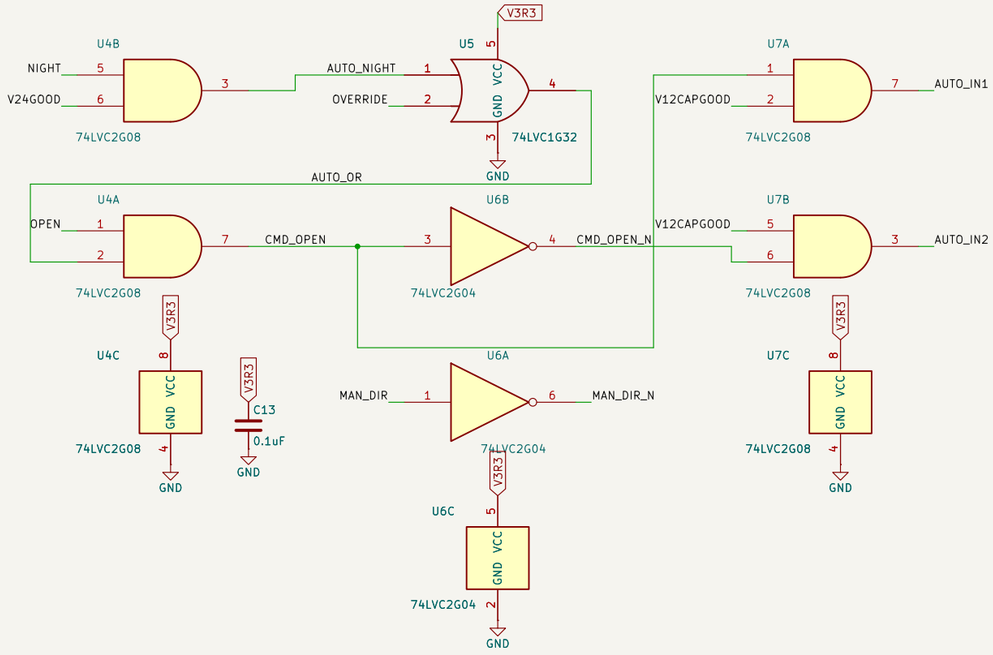


Note 3: U10 heatsink area for 1.5W peak:
suggest min 1 cm^2, prefer 12mm x 12mm or equiv.
pin 2 may be absent

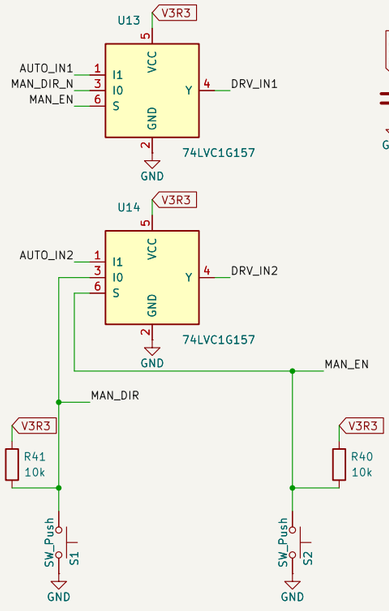
MOBO Connector (J1)



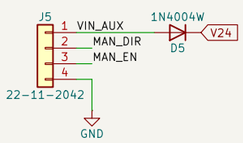
Control Logic (U4–U7)



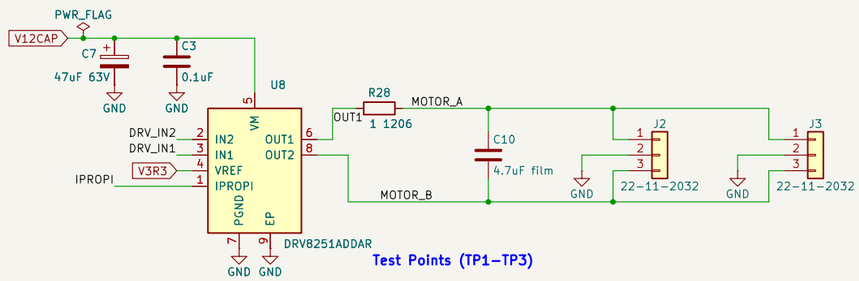
Manual/Auto Select (U13, U14)



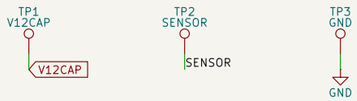
Aux Power / Manual (J5)



Motor Driver (U8, J2, J3)



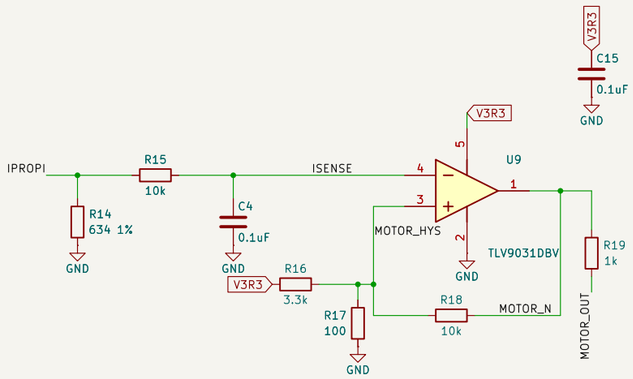
Test Points (TP1–TP3)



DRV8251A truth table

IN1	IN2	OUT1	OUT2
0	0	Hi-Z	Hi-Z
0	1	L	H
1	0	H	L
1	1	L	L

Current Sense (U9)

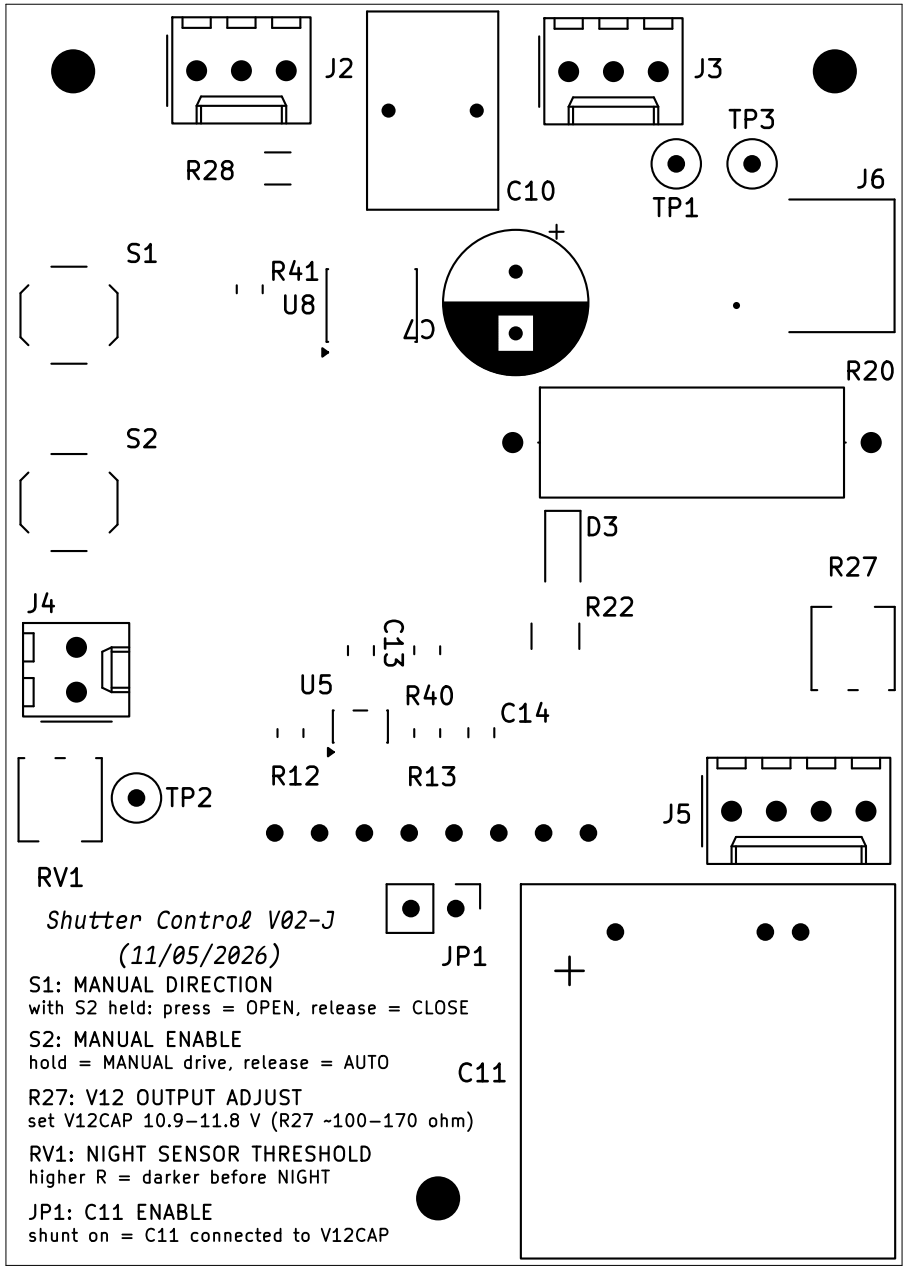


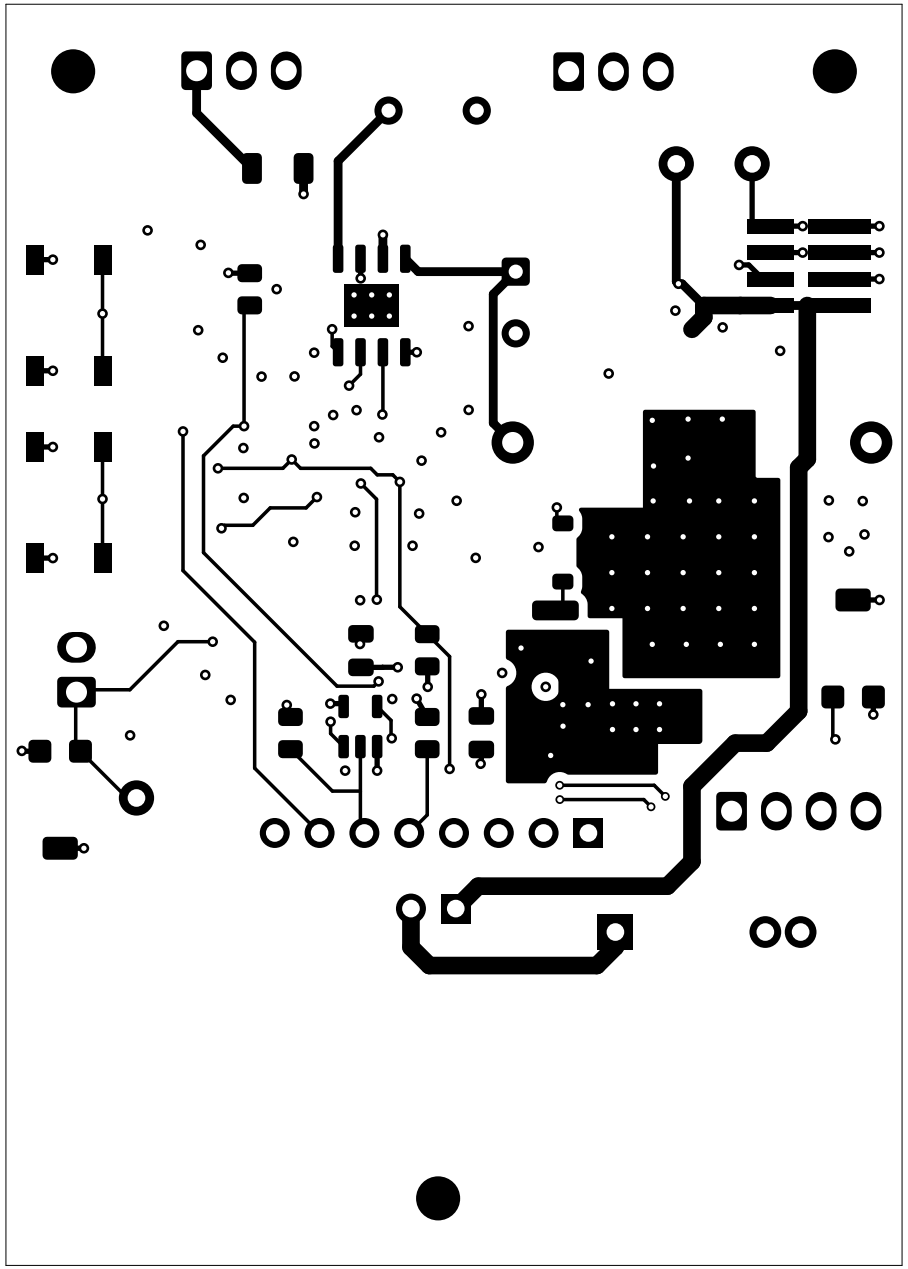
Note 2: IPROPI Iout = 1.58 mA/A; node internally clamped to V<=Vref;
motor current limits when V=Vref, here Ilim=3.3A

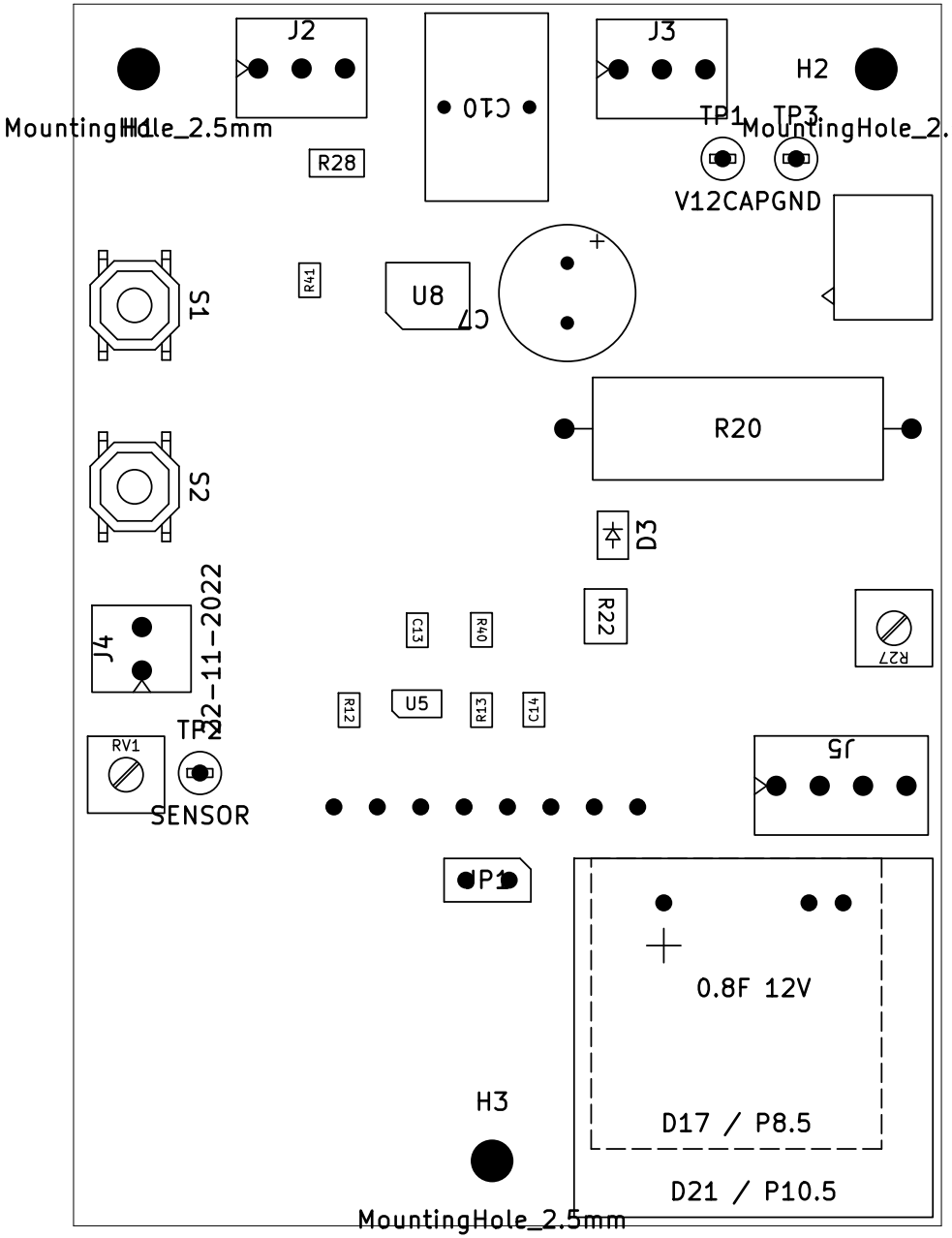
Sheet: /
File: shuttercontrol.kicad_sch

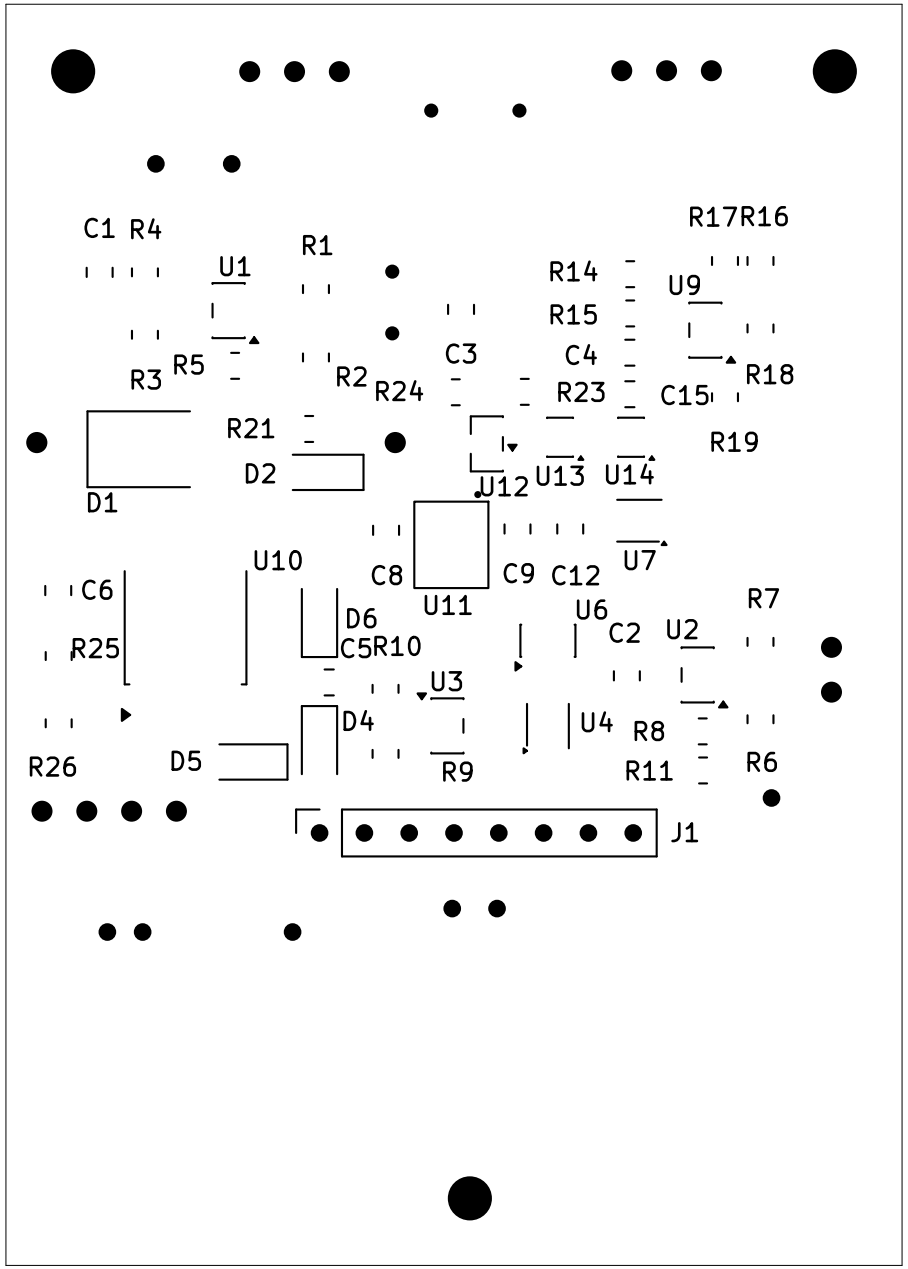
Title: New Shutter Controller

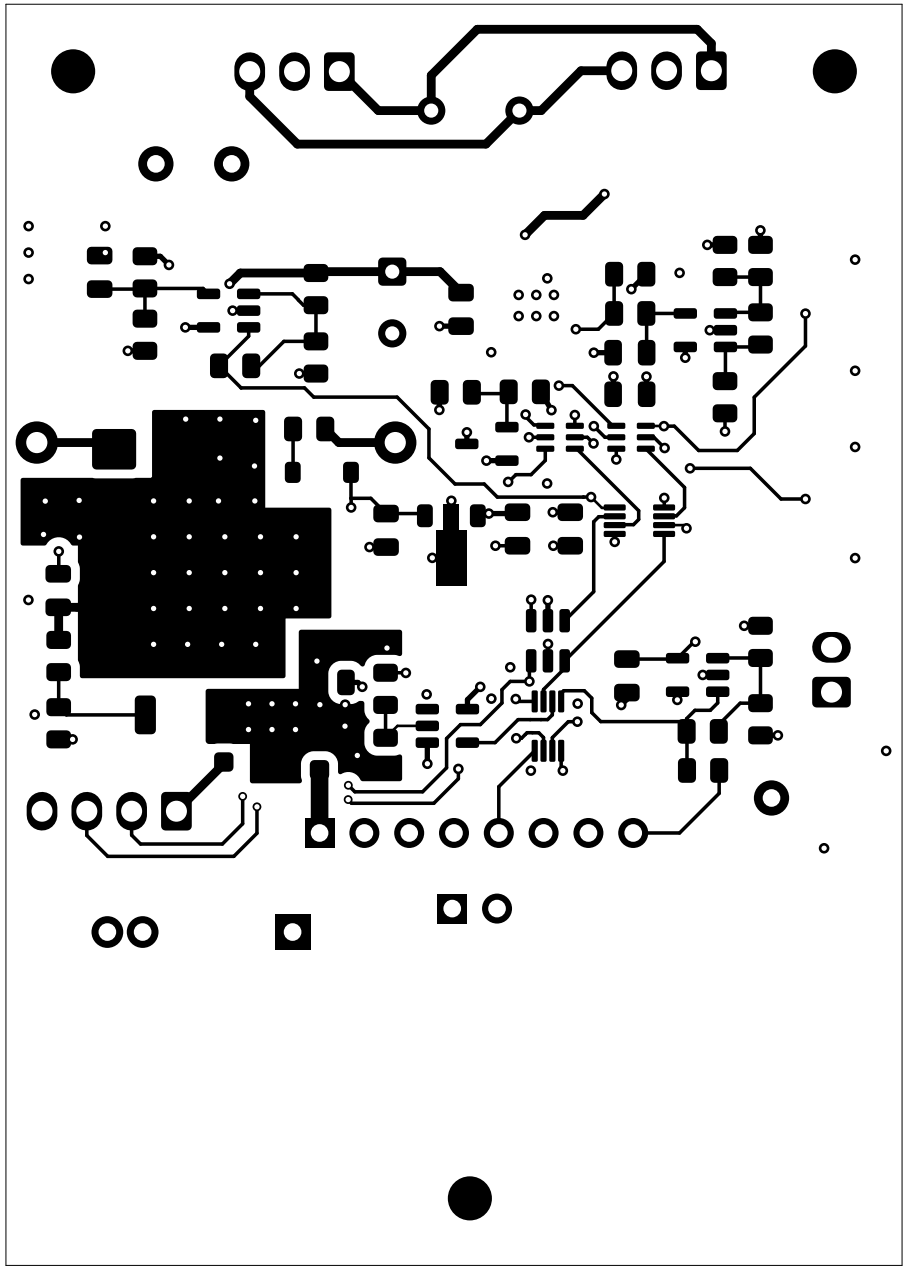
Size: A3	Date: 2026-11-05	Rev: v.02j
KiCad E.D.A. 10.0.8		Id: 1/1

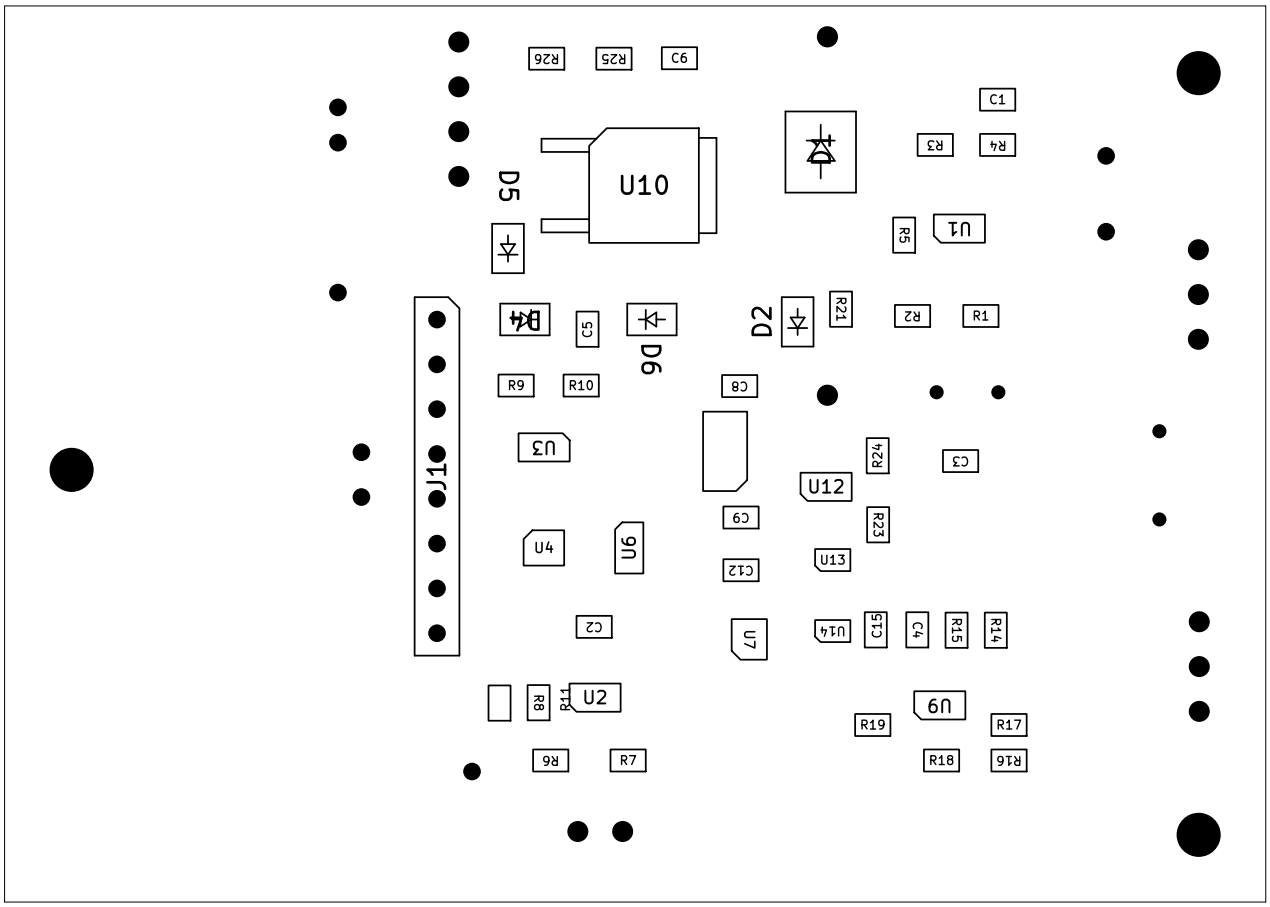

J2
J3
TP3
TP1
J6
R28
C10
S1
R41
U8
C7
R20
S2
D3
R27
J4
R22
C13
U5
R40
C14
R12
R13
TP2
J5
RV1
JP1
C11
Shutter Control V02-J
(11/05/2026)
S1: MANUAL DIRECTION
with S2 held: press = OPEN, release = CLOSE
S2: MANUAL ENABLE
hold = MANUAL drive, release = AUTO
R27: V12 OUTPUT ADJUST
set V12CAP 10.9–11.8 V (R27 ~100–170 ohm)
RV1: NIGHT SENSOR THRESHOLD
higher R = darker before NIGHT
JP1: C11 ENABLE
shunt on = C11 connected to V12CAP




J2
J3
H2
H1
MountingHole_2.5mm
TP1
TP3
MountingHole_2.5mm
R28
C10
V12CAPGND
S1
R41
U8
C7
R20
S2
D3
J4
22-11-2022
C13
R40
R22
R27
R12
U5
R13
C14
TP2
RV1
J5
SENSOR
JP1
0.8F 12V
H3
D17 / P8.5
D21 / P10.5
MountingHole_2.5mm


C1
R4
U1
R1
R17
R16
R14
U9
R15
C3
C4
R5
R3
R2
R24
R23
C15
R18
R21
R19
D2
U12
U13
U14
D1
U10
C8
C9
C12
U7
C6
U11
U6
R7
D6
C2
U2
R25
C5
R10
U3
D4
U4
R8
R26
D5
R9
R11
R6
J1




R26
R25
C6
C1
U10
R3
R4
D5
U1
R5
D4
C5
D6
D2
R21
R2
R1
R9
R10
C8
U3
J1
R24
C3
U12
C9
R23
U4
U6
U13
C12
C2
U7
U14
C15
C4
R15
R14
U2
R8
R11
U9
R19
R17
R6
R7
R18
R16